

The Cosmic Entropy Budget as Constraint on the EFC S -Field

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Abstract. Recent observational inventories of the cosmic entropy budget confirm that supermassive black holes (SMBHs) dominate the total entropy of the observable universe by many orders of magnitude. This working note examines the implications for Energy-Flow Cosmology (EFC), where the entropy field $S(\mathbf{x}, z)$ is a dynamical variable controlling gravitational coupling via $G_{\text{eff}} = G_N(1 + \mu(S))$. We show that the observed entropy hierarchy— $S_{\text{SMBH}} \gg S_{\text{stellar}} \gg S_{\text{CMB}}$ —maps naturally onto EFC’s regime structure: SMBHs occupy the $S \rightarrow S_{\text{max}}$ boundary (grid collapse in GRC ontology), while the CMB represents the $S \rightarrow S_{\text{max}}$ thermal limit from the opposite direction. We derive a consistency relation between the Bekenstein-Hawking entropy and EFC’s S -field normalization, identify testable predictions, and state the open gaps. This is a structural consistency note, not an empirical closure paper.

1 The Observational Entropy Inventory

The cosmic entropy budget has been quantified in detail by Egan & Lineweaver [1] and updated by Frampton & Kephart [2] and most recently by Chiang & Vieira [3]. The key results, expressed in Boltzmann units (k_B), are:

Table 1: Cosmic entropy budget (observable universe).

Component	S/k_B	Fraction
Supermassive black holes	$\sim 10^{104}$	$> 99.99\%$
Stellar-mass black holes	$\sim 10^{98}$	$< 0.01\%$
CMB photon bath	$\sim 10^{89}$	$\sim 10^{-15}$
Neutrino background	$\sim 10^{88}$	$\sim 10^{-16}$
Baryonic matter	$\sim 10^{81}$	$\sim 10^{-23}$

The dominance of SMBH entropy is overwhelming: a single $10^9 M_\odot$ black hole carries $S_{\text{BH}} \approx 10^{97} k_B$, via the Bekenstein-Hawking formula:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} = \frac{4\pi k_B G M^2}{\hbar c}, \quad (1)$$

where $A = 16\pi G^2 M^2 / c^4$ is the horizon area.

2 EFC Interpretation: The S -Field Hierarchy

2.1 The GRC ontological identification

In the GRC triad (Reality = Grid $\otimes$ Energy $\cdot$ Clarity(S)), different physical phenomena correspond to different S -configurations [4,5]:

Table 2: GRC triad configurations for astrophysical objects.

Phenomenon	S value	Clarity $C(S)$	Grid state
Particles	$S \rightarrow 0$	$C \rightarrow 1$	Sharp nodes
Fields	$S \approx 0.5 S_{\max}$	$C \approx 0.6$	Distributed flow
CMB bath	$S \rightarrow S_{\max}$	$C \rightarrow 0$	Near-thermal
Black holes	$S = S_{\max}$	$C = 0$	Grid collapse

The crucial structural point: both the CMB and black holes approach $S_{\max}$, but from *opposite physical directions*:

- The CMB is a *thermal* maximum: energy distributed uniformly across all modes, no structure, $C(S) \rightarrow 0$ because there is nothing to render sharply.
- SMBHs are a *gravitational* maximum: energy concentrated maximally, $S = S_{\max}$ via Bekenstein-Hawking, $C(S) \rightarrow 0$ because the grid itself collapses.

Both are entropy maxima. But the physical mechanisms are distinct, and EFC distinguishes them through the *path* in (E, S) space, not just the endpoint.

2.2 Mapping S_{BH} to the EFC S -field

The EFC entropy field $S(\mathbf{x}, z)$ is normalized to $[0, S_{\max}]$ where $S_{\max}$ corresponds to the regime boundary beyond which the L2 description breaks down. The natural identification is:

$$\frac{S_{\text{EFC}}}{S_{\max}} = \frac{S_{\text{BH}}}{S_{\text{BH},\max}(M)}, \quad (2)$$

where $S_{\text{BH},\max}(M)$ is the Bekenstein-Hawking entropy for a black hole of mass M . By definition, a Schwarzschild black hole saturates this bound: $S_{\text{EFC}} = S_{\max}$ for any actual BH.

This means that *every* SMBH in the EFC framework sits exactly at the grid-collapse boundary. The gravitational enhancement $\mu(S) \rightarrow \mu(S_{\max})$ reaches its limiting value, and the L2 description ceases to be valid inside the horizon.

3 Consistency Relations

3.1 The entropy production rate constraint

In EFC, the entropy production function $\Gamma(\rho)$ appears in the relativistic action [6]:

$$S_{\text{EFC}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{\text{Pl}}^2 F(\phi) R - \frac{1}{2} K(\rho) (\nabla \phi)^2 - V(\phi) - \lambda (\Box \phi - \Gamma(\rho)) \right]. \quad (3)$$

The entropy production in the vicinity of SMBHs is dominated by accretion. The accretion luminosity is:

$$L_{\text{acc}} = \eta \dot{M} c^2, \quad (4)$$

with radiative efficiency $\eta \approx 0.1$ for standard thin-disk accretion. The associated entropy production rate is:

$$\dot{S}_{\text{acc}} \approx \frac{L_{\text{acc}}}{T_{\text{acc}}}, \quad (5)$$

where $T_{\text{acc}} \sim 10^5\text{--}10^7$ K is the effective temperature of the accretion flow.

3.2 The SMBH thermostat conjecture

The observed entropy budget implies that SMBHs are the dominant entropy producers in the late universe. In EFC, this has a specific structural consequence: SMBH accretion *drives* the evolution of $S(z)$ at late times.

The $S(z)$ evolution in EFC is calibrated to reproduce DESI DR2 BAO [7] with a sigmoid transition at $z_t \approx 1$. The physical driver of this transition can now be identified: the epoch $z \approx 1\text{--}2$ is precisely when SMBH accretion peaks (the “quasar epoch”). As AGN activity declines at $z < 1$, entropy production slows, and the universe enters the regime where G_{eff} begins to differ measurably from G_N .

This suggests a **thermostat conjecture**: SMBHs regulate the cosmic S -field through their entropy production, and the transition in $S(z)$ at $z \sim 1$ is driven by the decline of AGN activity—not by an ad-hoc function.

Heuristic quantification: The comoving AGN bolometric luminosity density peaks at $\mathcal{L}_{\text{AGN}}(z) \approx 10^{40}$ W Mpc $^{-3}$ near $z \approx 2$ and declines by a factor ~ 10 to $z = 0$ [Hopkins et al. 2007]. The entropy production rate per comoving volume is $\dot{S}_{\text{AGN}} \sim \mathcal{L}_{\text{AGN}}/T_{\text{acc}}$. Integrating over $z = 0\text{--}2$ with $T_{\text{acc}} \sim 10^6$ K gives a cumulative ΔS that tracks the sigmoid shape of the calibrated $S(z)$. This is order-of-magnitude, not a fit—but it shows that the AGN luminosity function has the correct *shape* and *amplitude* to source the S -field transition.

4 Testable Predictions

- (P1) **SMBH density correlates with local μ .** Regions with higher SMBH number density should have higher local S and therefore *weaker* gravitational enhancement (μ closer to zero). This is testable via comparing RAR residuals in galaxies with and without active AGN.
- (P2) **$S(z)$ tracks AGN luminosity density.** The sigmoid $S(z)$ function should correlate with the observed AGN/quasar luminosity density evolution. The inflection point z_t should correspond to the peak of cosmic AGN activity ($z \approx 1-2$).
- (P3) **Void galaxies have lower local S .** Galaxies in cosmic voids (far from SMBHs) should show stronger gravitational enhancement, consistent with the RAR showing enhanced $g_{\text{obs}}/g_{\text{bar}}$ in low-density environments.
- (P4) **Black hole entropy scales with S_{max} .** If EFC's S_{max} corresponds to the Bekenstein-Hawking limit, then the normalization constant should satisfy:

$$S_{\text{max}} \sim \frac{4\pi G M_{\text{typical}}^2}{\hbar c}, \quad (6)$$

where M_{typical} is the characteristic mass scale of the dominant entropy producers (SMBHs with $M \sim 10^8-10^{10} M_{\odot}$).

5 Open Gaps

- (G1) **Quantitative $S(z)$ –AGN connection.** The thermostat conjecture (Section 3.2) is structurally motivated but not derived from the EFC action. A derivation requires computing $\Gamma(\rho)$ in SMBH-dominated environments and showing that the integrated entropy production reproduces the calibrated $S(z)$ sigmoid.
- (G2) **S -field normalization.** The mapping in Eq. (2) is a natural identification but not uniquely determined. The EFC S -field is dimensionless and normalized to $[0, S_{\max}]$; the Bekenstein-Hawking entropy is in Boltzmann units. The conversion factor requires specifying the grid-mode density of states, which is an L3 (quantum gravity) quantity not accessible within L2.
- (G3) **Interior description.** EFC’s L2 regime breaks down inside black hole horizons. The statement “ $S = S_{\max}$ at the horizon” is a boundary condition, not a full solution. The interior requires an L3 extension that does not yet exist.
- (G4) **Hawking radiation and S -field decay.** If SMBHs evaporate, does S decrease locally? In EFC, this would mean the grid “un-collapses” on Hawking timescales ($\sim 10^{67}$ yr for stellar-mass BHs). This is not observationally accessible but is required for theoretical self-consistency.

6 Summary

The observed cosmic entropy budget—dominated by supermassive black holes—is structurally consistent with EFC’s S -field framework:

- (1) SMBHs naturally sit at $S = S_{\max}$ (grid collapse boundary).
- (2) The CMB approaches $S_{\max}$ from the thermal direction.
- (3) The $S(z)$ transition at $z \sim 1$ plausibly tracks the decline of AGN entropy production (thermostat conjecture).
- (4) Void galaxies should show enhanced $\mu(S)$, testable with existing SPARC-type data.

This note establishes structural consistency between the observed entropy inventory and EFC’s dynamical S -field. It does not constitute a quantitative derivation of the entropy budget from first principles. Gaps G1 (AGN– $S(z)$ derivation) and G2 (S -field normalization) are the critical open problems.

Status: Working note. Closes the entropy-budget gap in the EFC ledger at the structural level.

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